Ziv Behar

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Education

University of California, Berkeley — ***May ’27***

B.S. Electrical Engineering and Computer Science (Honors) | *College of Engineering* — GPA: 3.98/4.0

Coursework: Efficient Data Structures & Algorithms, Machine Learning, Deep Neural Networks, Advanced Computer Architecture, Operating Systems, Robotics, Devices and Systems, Microelectronic devices.

Work Experience

AWS – *Software Development Engineer Intern* — ***May ’26 - Aug ’26***

- Migrated relational databases throttling at 1.5B+ reqs/day from an in-process system to a dedicated architecture scaling from ~50 to 500+ hosts, protecting pre/post-auth traffic for 13 request dimensions.
- Built a shadow-evaluation framework and automated rule-translation tooling to validate the new system against real-time production behavior and detect rule and decision drift.
- Optimized throttling handler and cache performance while building fail-safe behavior, production monitoring, and AI-agent-assisted CI/CD safeguards for safe rule deployment.

LIFTR App Store – *Software Engineer* — ***Nov ’25 - May ’26***

- Solely developed an acquired revenue-generating mobile fitness app using on-device ML to process exercise-form pose signals at 30 FPS and transform them into personalized real-time coaching feedback.
- Created a distinct LSTM-based workout recommendation engine that retrains incrementally as users log workouts, adapting recommendations over time.

Research

Berkeley Sensor and Actuator Center - *3 submissions to IEEE MEMS27* — ***Jan ’26 - Present***

- Invented the first physics-informed learned reconstruction method for in-air 3D ultrasonic imaging, using just 8 PMUT sensors (vs. hundreds conventionally) and cutting acquisition time by 10-100x while reconstructing target position, size, and orientation with 34 mm median surface error to ground truth.
- Developed methods for synthetic-aperture and non-line-of-sight PMUT imaging, improving resolution 11.8× to 2.87 mm and localizing occluded targets up to 1.42 m away within 0.9 mm of ground truth.

Projects

SnapSort — ***Jan ’25 - Dec ’25***

- Built a transformer-based event clustering system for large-scale personal camera rolls with 84% accuracy, including a reproducible data ingestion and labeling pipeline for unstructured photo streams.
- Built a mobile ↔ cloud ML pipeline for GPU-accelerated embedding generation and vector search using custom gRPC services and high-throughput parallelized inference.

MoffittStatus (acquired by ASUC) — ***Sep ’24 - Jan ’25***

- Launched a real-time library occupancy platform showing students where open seats are available within multiple UC Berkeley libraries, reaching 3.5K users in month one and 200k+ student views.
- Built the full stack from WiFi traffic sensor ingestion and processing to REST APIs and front-end.

Skills

Languages: Python, Java, JavaScript/TypeScript, C++, RISC-V, Swift, SQL.

Frameworks: PyTorch, CoreML, React/React Native, AWS, gRPC, Playwright, Logism, LTspice, PCBs.